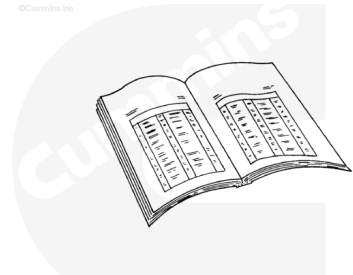


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.



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⚠ WARNING ⚠

A small amount of coolant can leak when servicing the coolant filter with the shutoff valve in the OFF position. To reduce the possibility of personal injury, avoid contact with hot coolant.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries and air starter to prevent accidental starting.
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Remove the alternator belt and alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Remove the fan belt tensioner assembly. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>).

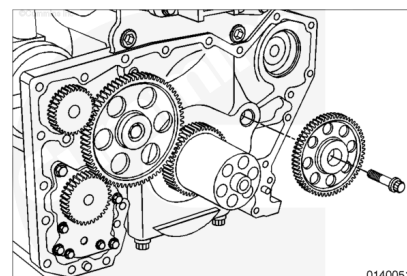
- Remove the vibration damper. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/89/89-001-052.html).
- Remove the alternator drive gear and shaft. Refer to Procedure 009-009 (/qs3/pubsys2/xml/en/procedures/89/89-009-009.html).
- Remove the front crankshaft seal. Refer to Procedure 001-023 (/qs3/pubsys2/xml/en/procedures/89/89-001-023.html).
- Remove the oil pan and related components. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html).
- Remove the block stiffener plate. Refer to Procedure 001-089 (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html).
- Remove the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html).

Remove

The bolt-in idler shafts have a flange that require the shaft, the idler gear, and the thrust washer to be removed as an assembly.

Remove the mounting capscrew from the idler gear.

Remove the gear assembly using two pry bars to gently pry the gear and the shaft from the block.



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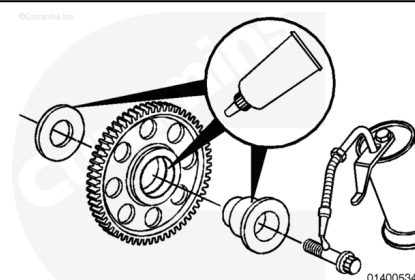
Install

⚠ CAUTION ⚠

The chamfer on the thrust washer must be turned toward the gear.

Use Lubriplate™ 105, or equivalent. Lubricate the gear bushing, shaft, and the thrust washer.

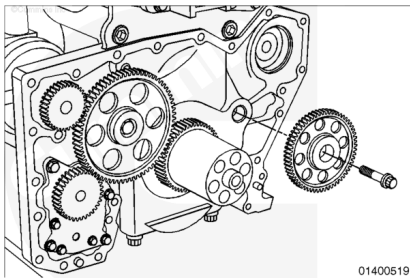
Use engine oil to lubricate the capscrew. Assemble the parts as shown.



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Use the capscrew to pull the shaft into the bore. Tighten the capscrew.

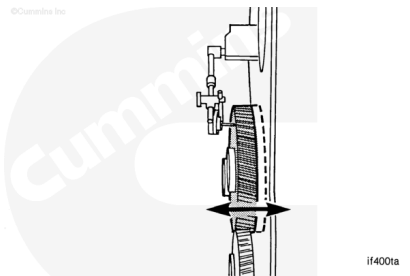
Torque Value: 277 n•m [204 ft-lb]



Use a dial indicator to measure the idler gear end clearance.

Idler Gear, Lubricating - Idler Gear End Clearance

mm		in
0.05	MIN	0.002
0.17	MAX	0.006



If the clearance is **not** within specifications, check for foreign material between the parts or check for proper location of the thrust washer. Oversize washers are available.

Check the idler gear backlash. Refer to Procedure 001-055 (</qs3/pubsys2/xml/en/procedures/89/89-001-055.html>).

Finishing Steps

- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>).
- Install the alternator drive gear and shaft. Refer to Procedure 009-009 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>).
- Install the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>).
- Install the block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>).
- Install the fan belt tensioner assembly. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>).



- Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>).
- Install the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>).
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>).
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>).
- Install the oil pan and related components. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Install new oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/89/89-007-013.html>).
- Connect the batteries.
- Operate the engine to normal operating temperature. Check for leaks.